

Week 7, Lecture 1: How Sure Are We? Uncertainty from the Shape of the Likelihood

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** Last week we found the single best-supported parameter value (the MLE). A point estimate alone doesn't tell us much. Today's question: how uncertain is a parameter estimate, and how do we turn that uncertainty into a confidence interval? We'll also take a first, brief look at the Bayesian alternative.
- **Read:** Bolker Ch. 6 through section 6.5 (printed p. 227-267; stop at 6.6, "Comparing models"). Focus on 6.4 (p. 249-260) and 6.5 (p. 260-267). Skim the Bayesian parts (6.2.2, p. 237-243, and 6.4.2, p. 258-260) for the main idea; Wednesday's lecture goes into Bayesian inference in more detail. Bring questions about the reading.
- **Review:** the Poisson distribution and its likelihood (last week's pika haypile example); log rules; derivatives, including second derivatives and the derivative of $\log x$ and of $1/x$.
- **Optional:** look up what a chi-squared (χ^2) distribution is. You don't need its formula, only a rough idea of its shape.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

Beat	Time
WOD: <i>Reading the Interval</i> (in class)	~8 min
Lecture	~35 min
Discussion	~7 min

Source material: Bolker 6.4-6.5 (the whole lecture follows his logic), 6.2.2 for the Bayesian aside; legacy `LECTURE4.Rmd` ("Estimating parameter uncertainty" through "The usual way to generate CIs") for the slice/profile/Hessian sequence; legacy `LECTURE6.Rmd` for the prior $\times$ likelihood picture.

Example choice. The one-parameter thread reuses last Monday's pika haypile data (Week 6, Lecture 1), so students see a familiar likelihood curve and only the new idea (its *width*) is

unfamiliar. The numbers are friendly: the curvature at the MLE is 2, so the standard error is $1/\sqrt{2}$. The two-parameter piece uses Bolker's tadpole predation data (Figures 6.7-6.8, in the assigned reading). I avoided the frog chorus data from Lab 5 Act 2 on purpose: if Act 2 went home as a take-home, showing its fitted $\hat{a}, \hat{b}$ on Monday would give away 2b before it's due.

Where this sits in the week

- **Last week:** likelihood, the NLL, and the MLE (Monday); Lab 5 found an MLE by calculus (Act 1) and with `optim()` (Act 2). Lab 5 Act 1 Part 3 already showed two NLL curves, 3 vs. 30 wetlands, with the same $\hat{p} = 0.4$ but different widths, and asked which survey says more about p . Today formalizes that answer.
- **Today:** the width and curvature of the NLL curve become a confidence interval, two ways (likelihood ratio, and the quadratic or Hessian approximation). Brief Bayesian contrast.
- **Tomorrow (Lab 6):** Bliss's beetle fumigation data. Act 1 (paper) draws a slice and a profile on a very tilted (a, b) surface and uses a 2×2 Hessian to get an SE by hand. Act 2 fits the LD50 directly and compares Hessian and profile intervals. Section 7 below is the preview, so keep it visual and don't work the beetle numbers.
- **Wednesday:** Bayesian inference in more detail (optimization moved to next week). Wednesday's take-home WOD, *Curvature by Hand*, has students derive the Poisson and binomial curvatures themselves, so today should show the curvature value (2) but **not** derive it on the board.

WOD: Reading the Interval (~8 min)

File: `wod/WOD_Week7_Lecture1_ReadTheInterval.Rmd`. Deliberately light (no algebra), since students have had several math-heavy WODs in a row. It gives students the "2 units" rule of thumb *before* the lecture explains it, so section 5 can pay it off.

Chytrid swabs: 2 positives in 20 frogs, binomial, printed NLL-minus-min curve. About 4 min alone, 2 min with a neighbor, 2 min debrief.

- **1a.** $\hat{p} = 0.10$.
- **1b.** Within 2 units: about 0.02 to 0.28 (exact: 0.017 to 0.282).
- **1c.** Not symmetric: 0.08 below $\hat{p}$, 0.18 above. The curve shoots up near 0 because prevalence can't be negative and 2 positives already rule out values very close to 0; high values are ruled out more gradually.
- **2a.** 8 of 80: same $\hat{p} = 0.10$, curve 4 times steeper, range about 0.05 to 0.18 (narrower, and less lopsided).

Debrief lightly: confirm 1a and 1b, take one answer to 1c, then say "the rest of today explains where the 2 comes from and why the curve's width matters." Save the full asymmetry point for section 5, and don't mention the Wald interval's negative lower bound for these data: that's the payoff of 2d in Wednesday's take-home.

Part 1: Lecture content

1. Big picture (~3 min)

Put the loop on the board:

THEORY $\rightarrow$ MODEL(S) $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat

- So far in the “fit” step we have produced one number per parameter, the MLE. Bolker (p. 249): point estimates are useless without a measure of uncertainty.
- Today’s question: *which parameter values are still plausible given the data?* Answer: all the information is in the shape of the likelihood curve near its peak. A narrow valley means the data rule out most values; a broad valley means they don’t.
- The same likelihood feeds two schools of inference. Frequentists use it to build confidence intervals (today’s focus). Bayesians combine it with a prior to get a probability distribution for the parameter (brief look today, more on Wednesday).
- Popperian tie-in, worth saying out loud: a confidence interval is the set of parameter values the data *fail to reject*. Values outside it are falsified (at the 5% level); values inside survive. None is proven, including the MLE.

2. Questions about the reading (~4 min)

Open the floor. If nobody bites, seed with one of these:

- “Bolker says slices are ‘usually wrong.’ Why would fixing the other parameters at their best values be a problem?” (Answer comes in section 7.)
- “Why does doubling the data make the *maximum* likelihood smaller but the estimate better?” (Bolker’s coin footnote, p. 254: 1 head in 2 flips has probability 0.5; 500 in 1000 has probability 0.025. That doesn’t make the coin less fair.)
- “Where does the 1.92 come from?” (Section 5.)

Don’t answer at length now if the question is covered later; say “hold that, it’s coming in ten minutes” and write it in the corner of the board so you remember to close the loop.

3. Bayesian inference, in five minutes (~4 min)

Keep this conceptual. One equation, one picture.

$$\underbrace{P(\lambda \mid y)}_{\text{posterior}} = \frac{\mathcal{L}_6(\lambda) P(\lambda)}{\int \mathcal{L}_6(\lambda) P(\lambda) d\lambda} \propto \text{likelihood} \times \text{prior}$$

- The likelihood is *not* a probability distribution over λ (Week 5 WOD, question 2d). Multiplying by a prior and dividing by the total area turns it into one: the posterior.

- The denominator is just a constant that makes the area equal 1. In practice it is hard to compute, which is why MCMC exists (later in the semester). Wednesday picks this up.
- With a flat prior, the posterior has the same shape as the likelihood. An informative prior pulls it toward the prior.

```
haypiles <- c(2, 5, 3, 0, 4, 4)
n_plots <- length(haypiles)
lambda_hat <- mean(haypiles)

nll_pois <- function(lambda, y) {
  -sum(dpois(y, lambda = lambda, log = TRUE))
}

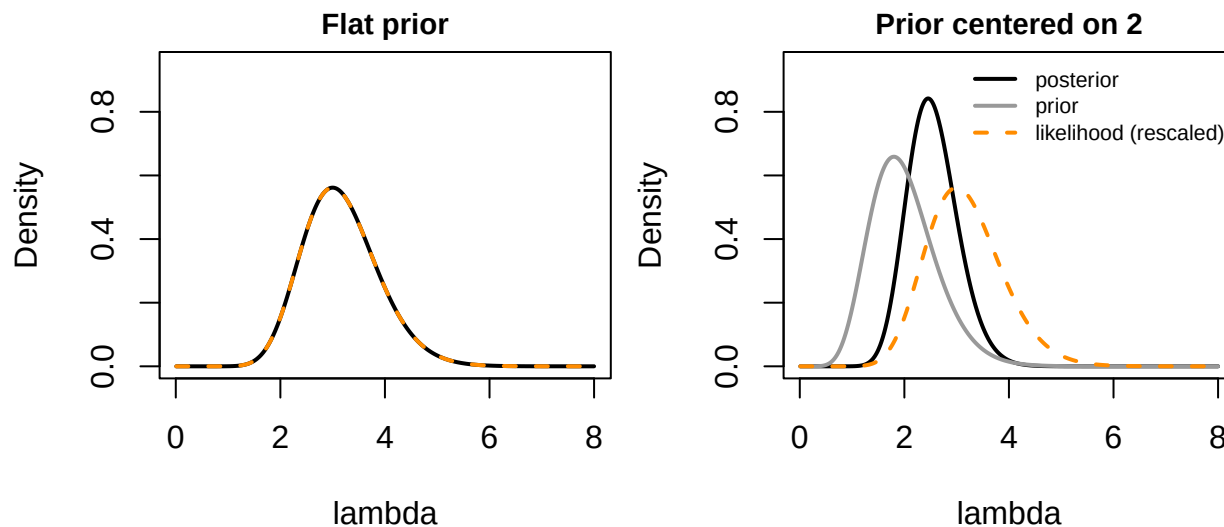
lambda_grid <- seq(0.01, 8, by = 0.01)
lik_vals <- sapply(lambda_grid, function(l) prod(dpois(haypiles, l)))
```

For the informative prior, suppose an earlier survey at a nearby talus field suggests about 2 haypiles per plot, encoded as a Gamma prior with mean 2 (shape 10, rate 5). Everything below is brute force on a grid, which is what legacy LECTURE6.Rmd did and is easier to explain than conjugacy.

```
step <- diff(lambda_grid)[1]
post_from_prior <- function(prior_vals) {
  numer <- lik_vals * prior_vals
  numer / sum(numer * step)
}

prior_flat <- rep(1, length(lambda_grid))
prior_info <- dgamma(lambda_grid, shape = 10, rate = 5)
post_flat <- post_from_prior(prior_flat)
post_info <- post_from_prior(prior_info)
lik_scaled <- lik_vals / max(lik_vals) * max(post_flat)

par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
plot(lambda_grid, post_flat, type = "l", lwd = 2, ylim = c(0, 0.95),
      xlab = "lambda", ylab = "Density", main = "Flat prior", cex.main = 0.9)
lines(lambda_grid, lik_scaled, lwd = 2, lty = 2, col = "darkorange")
plot(lambda_grid, post_info, type = "l", lwd = 2, ylim = c(0, 0.95),
      xlab = "lambda", ylab = "Density", main = "Prior centered on 2",
      cex.main = 0.9)
lines(lambda_grid, prior_info, lwd = 2, col = "grey60")
lines(lambda_grid, lik_scaled, lwd = 2, lty = 2, col = "darkorange")
legend("topright", c("posterior", "prior", "likelihood (rescaled)"),
      lwd = 2, lty = c(1, 1, 2), col = c("black", "grey60", "darkorange"),
      bty = "n", cex = 0.7)
```



In the left panel the posterior and the rescaled likelihood sit on top of each other. In the right panel the posterior mode moves from 3 to 2.45, between the prior and the data. (The likelihood is rescaled only so it fits on the same axes; its own vertical scale means nothing, as last Monday’s sticking point said.)

The payoff for Bayesians: a 95% **credible interval** is read straight off the posterior, and it *is* a probability statement about λ . With the flat prior the posterior is `Gamma(shape 19, rate 6)`, so the equal-tailed interval is:

```
cred_int <- qgamma(c(0.025, 0.975), shape = sum(haypiles) + 1, rate = n_plots)
round(cred_int, 2)
```

```
## [1] 1.91 4.74
```

Hold on to these numbers. They get compared with the frequentist interval at the end of section 5.

Sticking point: students hear “prior” and think “bias.” Say plainly that with a flat prior and a reasonable amount of data the Bayesian and likelihood answers nearly coincide; the frameworks differ more in *interpretation* than in numbers for simple problems like this.

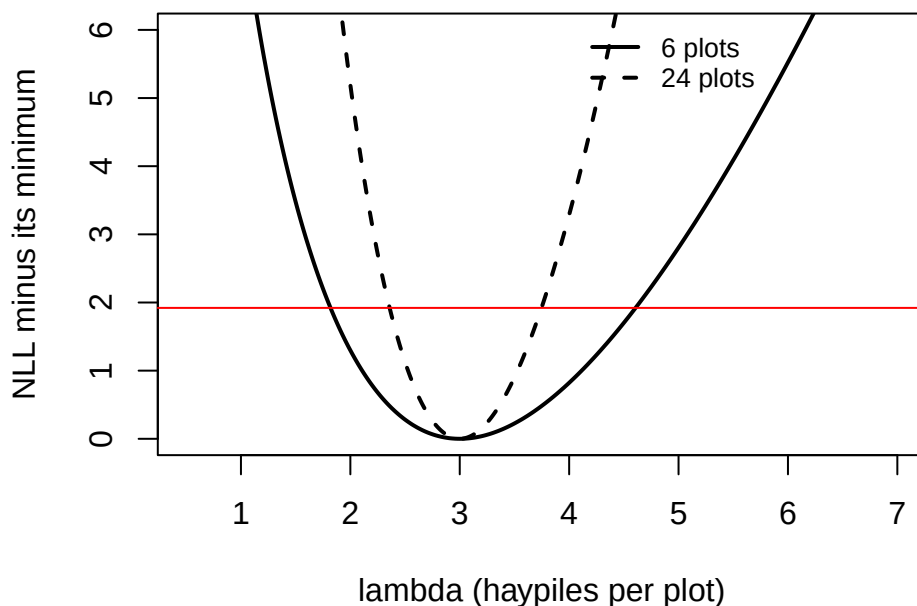
4. Curvature is information (~6 min)

Callback: Lab 5 Act 1, question 3a. Same $\hat{p}$, but the 30-wetland curve was narrower. Now do the same thing with the pikas. Imagine the biologist had surveyed 24 plots, with the same mean (each of the six counts appearing four times).

```
haypiles_24 <- rep(haypiles, 4)
dnll_6 <- sapply(lambda_grid, nll_pois, y = haypiles) - nll_pois(3, haypiles)
dnll_24 <- sapply(lambda_grid, nll_pois, y = haypiles_24) - nll_pois(3, haypiles_24)
lr_cut <- qchisq(0.95, df = 1) / 2

par(mar = c(4, 4.5, 0.5, 0.5))
```

```
plot(lambda_grid, dnll_6, type = "l", lwd = 2, xlim = c(0.5, 7), ylim = c(0, 6),
     xlab = "lambda (haypiles per plot)", ylab = "NLL minus its minimum")
lines(lambda_grid, dnll_24, lwd = 2, lty = 2)
abline(h = lr_cut, col = "red")
legend(x = 4.0, y = 6.2, c("6 plots", "24 plots"), lwd = 2, lty = c(1, 2),
     bty = "n", cex = 0.8)
```



- Both bottom out at $\lambda = 3$.
- The NLL is a *sum* over observations, so 4 times the data means 4 times the NLL, everywhere. The valley gets 4 times steeper (Bolker p. 254).
- Steeper valley = the fit gets worse faster as you move away from the MLE = the data rule out more values = a narrower interval.

Sticking point: the *height* of the minimum also changes with n (more data means a smaller maximum likelihood, larger minimum NLL). That's why the figure subtracts the minimum. Only *differences* in NLL between parameter values on the same data mean anything.

5. The likelihood ratio interval (~7 min)

We need a rule for “how far up the valley wall is still plausible.”

- **Informal version (Edwards, cited in Bolker p. 254):** this is the WOD's rule of thumb. Keep every value within 2 log-likelihood units of the best. That's any value whose likelihood is at least $1/e^2 \approx 1/7.4$ of the maximum.
- **Frequentist version (the likelihood ratio test):** the deviance, $2 [\text{NLL}(\lambda) - \text{NLL}(\hat{\lambda})]$, has approximately a χ^2 distribution with 1 degree of freedom when λ is the true value. Its 95th percentile is 3.84, so the cutoff on the NLL scale is $3.84/2 = 1.92$. (This is Wilks' theorem. Don't derive it; Bolker's footnote on p. 255 gives the intuition, which

you can paraphrase: near the MLE the NLL is a parabola, and a squared normal deviate is χ^2_1 .)

- The 95% interval is every λ whose NLL is within 1.92 of the minimum: where the curve crosses the red line in the figure above.

```
lr_cut

## [1] 1.920729

f_cut <- function(l, y) nll_pois(l, y) - nll_pois(mean(y), y) - lr_cut
lr_int <- c(uniroot(f_cut, c(0.5, 3), y = haypiles)$root,
           uniroot(f_cut, c(3, 8), y = haypiles)$root)
lr_int_24 <- c(uniroot(f_cut, c(1, 3), y = haypiles_24)$root,
              uniroot(f_cut, c(3, 6), y = haypiles_24)$root)
round(rbind(plots_6 = lr_int, plots_24 = lr_int_24), 2)

##           [,1] [,2]
## plots_6  1.82 4.61
## plots_24 2.36 3.75
```

- With 6 plots: $\lambda \in [1.82, 4.61]$. With 24: $[2.36, 3.75]$, about half as wide (4 times the data, $\sqrt{4} = 2$ times narrower).
- The 6-plot interval is **not symmetric** around 3 (it reaches 1.61 above but only 1.18 below). That's real: counts can't go below zero, and the NLL curve is steeper on the low side. Keep this in mind for section 6.
- WOD callback: the chytrid curve was lopsided for the same reason, and more so, since $\hat{p} = 0.1$ sits closer to its boundary.
- Popperian reading: $\lambda = 1.5$ is falsified at the 5% level; $\lambda = 2$ survives; so does 4. The interval is the set of hypotheses that survive.

Now compare with the Bayesian credible interval from section 3:

Method	95% interval
Likelihood ratio	[1.82, 4.61]
Bayesian, flat prior	[1.91, 4.74]

Close numbers, different meanings. The confidence interval is a statement about the *procedure*: if we repeated the survey many times, about 95% of intervals built this way would contain the true λ . The credible interval is a statement about λ itself, given the data and the prior. This distinction feeds the discussion.

6. The quadratic approximation and the Hessian (~6 min)

The LR interval needs a search (here `uniroot()`; with many parameters, a full profile, section 7). The shortcut every canned routine uses: near its minimum, the NLL looks like a parabola.

A Taylor expansion around $\hat{\lambda}$ (Calc II), using the fact that the first derivative is zero at the minimum:

$$\text{NLL}(\lambda) \approx \text{NLL}(\hat{\lambda}) + \frac{1}{2} \text{NLL}''(\hat{\lambda}) (\lambda - \hat{\lambda})^2$$

Set the increase equal to the LR cutoff and solve:

$$\frac{1}{2} \text{NLL}''(\hat{\lambda}) (\lambda - \hat{\lambda})^2 = 1.92 \Rightarrow \lambda = \hat{\lambda} \pm \frac{1.96}{\sqrt{\text{NLL}''(\hat{\lambda})}}$$

($\sqrt{3.84} = 1.96$, the familiar normal quantile.) So the **standard error** is $1/\sqrt{\text{curvature}}$. Sharp valley, big second derivative, small SE. This is Bolker's eq. 6.5.5.

For the pika data the curvature at $\hat{\lambda} = 3$ is 2 (from `optim()` below). **Don't derive it on the board:** working out NLL'' for the Poisson and binomial is Wednesday's take-home WOD.

```
fit <- optim(par = 2, fn = nll_pois, y = haypiles, method = "Brent",
            lower = 0.01, upper = 20, hessian = TRUE)
fit$hessian                # curvature at the MLE: 2
```

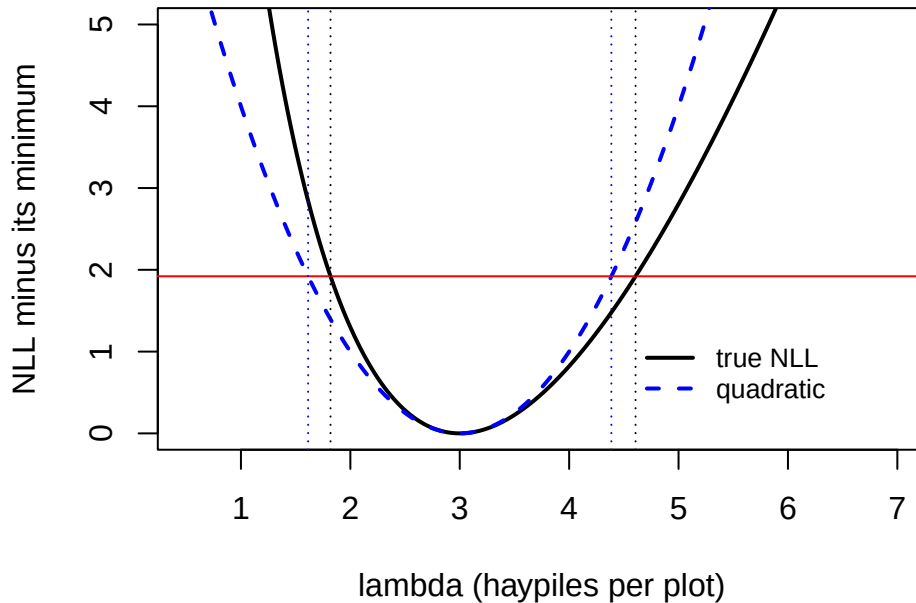
```
##      [,1]
## [1,]    2
```

```
se_lambda <- 1 / sqrt(fit$hessian[1, 1])
wald_int <- fit$par + c(-1, 1) * qnorm(0.975) * se_lambda
round(c(se = se_lambda, lower = wald_int[1], upper = wald_int[2]), 2)
```

```
##      se lower upper
## 0.71  1.61  4.39
```

`hessian = TRUE` asks `optim()` for the second derivative, computed numerically by finite differences. Nobody has to do the calculus.

```
quad_approx <- 0.5 * fit$hessian[1, 1] * (lambda_grid - lambda_hat)^2
par(mar = c(4, 4.5, 0.5, 0.5))
plot(lambda_grid, dnll_6, type = "l", lwd = 2, xlim = c(0.5, 7), ylim = c(0, 5),
     xlab = "lambda (haypiles per plot)", ylab = "NLL minus its minimum")
lines(lambda_grid, quad_approx, lwd = 2, lty = 2, col = "blue")
abline(h = lr_cut, col = "red")
abline(v = lr_int, lty = 3)
abline(v = wald_int, lty = 3, col = "blue")
legend(x = 4.5, y = 1.3, c("true NLL", "quadratic"), lwd = 2, lty = c(1, 2), cex = 0.8,
     col = c("black", "blue"), bty = "n")
```

- Quadratic (Wald) interval: $[1.61, 4.39]$, symmetric by construction.
- LR interval: $[1.82, 4.61]$, shifted right, following the real curve.
- With 6 plots the parabola misses the curve's lopsidedness. With more data the NLL gets closer to a parabola and the two agree (both are large-sample approximations; the LR one is usually the better of the two).

Sticking point: why does a *second* derivative measure uncertainty? The first derivative is zero at the MLE by definition, so it carries no information about width. The second derivative says how fast the slope picks up as you walk away, which is how fast the fit degrades. Draw two parabolas with the same vertex, one narrow and one wide, and ask which has the bigger second derivative.

7. More than one parameter: slices, profiles, the Hessian matrix (~4 min)

Keep this mostly visual. Bolker's tadpole data (Figures 6.7-6.8): Holling type II predation, binomial number killed, $p = a/(1 + ahN)$.

```
library(emdbook)
data(ReedfrogFuncresp)
initial_n <- ReedfrogFuncresp$Initial
killed <- ReedfrogFuncresp$Killed

nll_holling <- function(a, h) {
  p <- a / (1 + a * h * initial_n)
  -sum(dbinom(killed, size = initial_n, prob = p, log = TRUE))
}
```

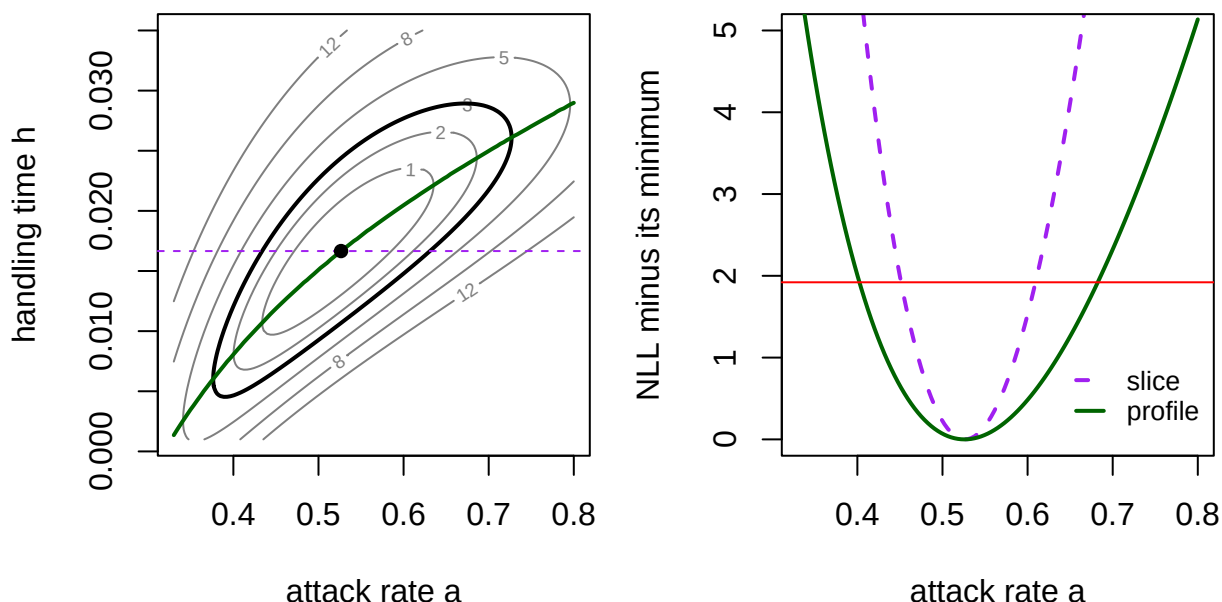
```
tad_fit <- optim(c(0.5, 0.0125), function(par) nll_holling(par[1], par[2]),
  method = "L-BFGS-B", lower = c(0.01, 0.0001),
  upper = c(2, 0.1), hessian = TRUE)
a_hat <- tad_fit$par[1]
h_hat <- tad_fit$par[2]
round(tad_fit$par, 4)
```

```
## [1] 0.5265 0.0167
```

```
a_seq <- seq(0.33, 0.8, length.out = 121)
h_seq <- seq(0.001, 0.035, length.out = 121)
nll_surf <- outer(a_seq, h_seq, Vectorize(nll_holling))
min_nll <- tad_fit$value

profile_h <- sapply(a_seq, function(a) {
  optimize(function(h) nll_holling(a, h), c(0.0001, 0.1))$minimum
})
profile_a <- sapply(seq_along(a_seq), function(i) nll_holling(a_seq[i], profile_h[i]))
slice_a <- sapply(a_seq, nll_holling, h = h_hat)

par(mfrow = c(1, 2), mar = c(4, 4.5, 0.5, 0.5))
contour(a_seq, h_seq, nll_surf - min_nll, levels = c(1, 2, 3, 5, 8, 12),
  xlab = "attack rate a", ylab = "handling time h", col = "grey50")
contour(a_seq, h_seq, nll_surf - min_nll, levels = qchisq(0.95, 2) / 2,
  add = TRUE, lwd = 2, drawlabels = FALSE)
abline(h = h_hat, lty = 2, col = "purple")
lines(a_seq, profile_h, lty = 1, lwd = 2, col = "darkgreen")
points(a_hat, h_hat, pch = 16)
plot(a_seq, slice_a - min_nll, type = "l", lwd = 2, lty = 2, col = "purple",
  ylim = c(0, 5), xlab = "attack rate a", ylab = "NLL minus its minimum")
lines(a_seq, profile_a - min_nll, lwd = 2, col = "darkgreen")
abline(h = lr_cut, col = "red")
legend(x = 0.63, y = 1.1, c("slice", "profile"), lwd = 2, lty = c(2, 1), cex = 0.8, seg.
  col = c("purple", "darkgreen"), bty = "n")
```



- **Left:** the NLL surface. The valley is a tilted ellipse: estimates of a and h are correlated (a higher attack rate can be offset by a longer handling time). Thick contour: 95% joint region (χ^2_2 cutoff, $5.99/2 \approx 3$).
- **Slice** (purple dashed): hold h at $\hat{h}$, vary a . **Profile** (green): for each a , re-optimize h . The profile follows the floor of the tilted valley.
- **Right:** the slice is steeper, so it gives a falsely narrow interval. It ignores that h can adjust to compensate. That's why Bolker calls slices “usually wrong.” The profile is the right way to get an LR interval for one parameter.

Profiles cost one optimization per grid point. The quadratic shortcut again: invert the Hessian matrix to get the variance-covariance matrix.

```
vcov_tad <- solve(tad_fit$hessian)
round(sqrt(diag(vcov_tad)), 4)      # standard errors of a and h

## [1] 0.0710 0.0049

round(cov2cor(vcov_tad), 2)        # correlation between a-hat and h-hat

##      [,1] [,2]
## [1,] 1.00 0.82
## [2,] 0.82 1.00
```

The off-diagonal correlation (about 0.82) is the tilt of the ellipse, in one number. Don't go deeper into matrix inversion today; the point is that the same “curvature = precision” idea scales up.

8. Return to the big picture (~1 min)

- The MLE says where the valley bottom is; the curvature says how wide it is. Width = uncertainty.
- Three frequentist routes to an interval: LR on the curve (or profile), quadratic/Hessian (fast, symmetric, large-sample), and bootstrapping (not covered today). One Bayesian route: prior $\times$ likelihood, read the interval off the posterior.
- Next in the loop: COMPARE. The likelihood ratio test we used today to build an interval is the same test we'll use to compare models.

Discussion (~7 min)

Pick one or two:

1. Our two 95% intervals for λ are about [1.8, 4.6] (likelihood ratio) and [1.9, 4.7] (Bayesian, flat prior). A land manager asks, “So there’s a 95% chance the true density is between 1.8 and 4.6?” Which interval lets you say yes? What would you say instead for the other one?
2. Is $\lambda = 1.5$ haypiles per plot *disproven* by these data? Is $\lambda = 3$ *proven*? What would change your answer?
3. A reviewer says six plots is too few to trust any interval. Of the intervals we built today, which would you trust most with $n = 6$, and why? What would you do to check?

Question 1 is the one most likely to surface the confidence vs. credible misconception. Question 2 reinforces the survive/falsify vocabulary.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

WOD ~8 min. Lecture ~35 min. Discussion ~7 min.

WOD: Reading the Interval (~8 min)

Chytrid: 2 of 20 positive. 4 min alone, 2 with neighbor.

- 1a: $\hat{p} = 0.10$
- 1b: about 0.02 to 0.28
- 1c: lopsided (0.08 below, 0.18 above); boundary at 0
- 2a: 8 of 80, same $\hat{p}$, narrower (about 0.05 to 0.18)
- Don't mention the Wald interval here (take-home 2d)

1. Big picture (~3 min)

- Loop on board: THEORY $\rightarrow$ MODEL $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat
- So far: MLE = one number. Useless without uncertainty (Bolker p. 249)
- Today: **shape of the likelihood = how sure we are**
- Same likelihood, two schools: CI (today), posterior (brief; more Wed.)
- CI = values that *survive*; outside = falsified; nothing proven

2. Reading questions (~4 min)

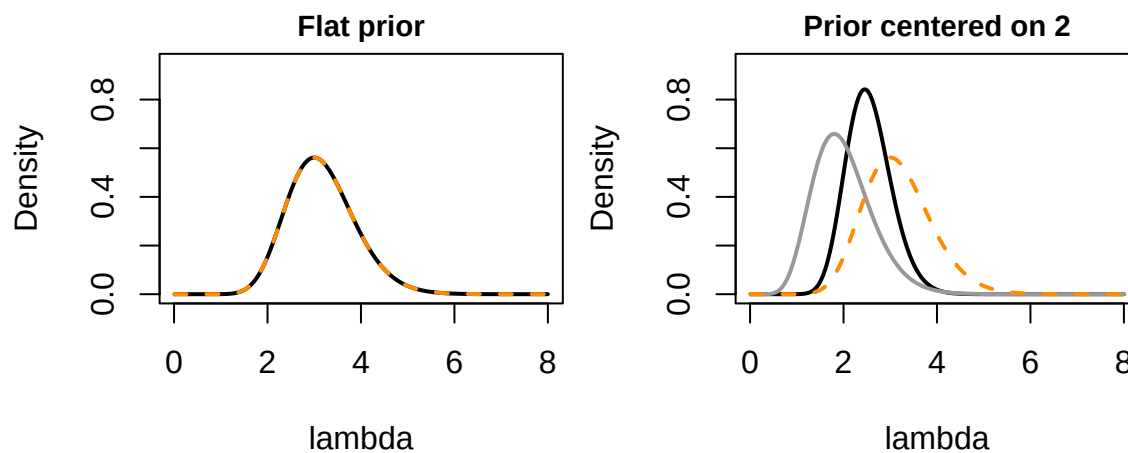
Seeds if silent:

- Why are slices “usually wrong”? (sec. 7)
- More data, *smaller* max likelihood, better estimate? (coin footnote)
- Where does 1.92 come from? (sec. 5)

Park answers on the board corner.

3. Bayes in five minutes (~4 min)

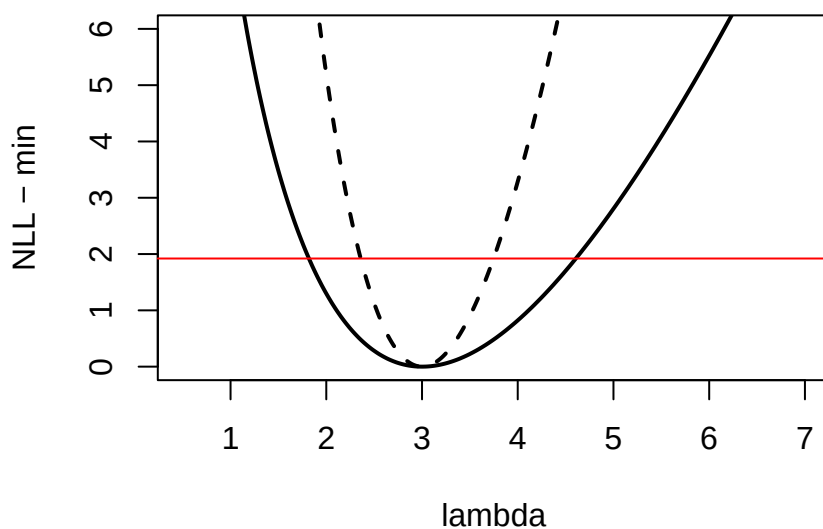
$$P(\lambda \mid y) = \frac{\mathcal{L}_6(\lambda) P(\lambda)}{\int \mathcal{L}_6(\lambda) P(\lambda) d\lambda}$$



- Likelihood $\times$ prior, rescaled to area 1
- Flat prior: posterior shape = likelihood shape
- Denominator is hard: why MCMC exists (later)
- Credible interval: $[1.91, 4.74]$. Hold for sec. 5

4. Curvature is information (~6 min)

Pikas: 2, 5, 3, 0, 4, 4. Same data $\times 4 = 24$ plots.



- Callback: Lab 5 Act 1, 3a (3 vs 30 wetlands)
- NLL is a sum: $4 \times \text{data} = 4 \times \text{steeper}$
- **Sticking point:** height of minimum changes with n ; only *differences* matter

5. Likelihood ratio interval (~7 min)

- Edwards: within 2 log-lik units (the WOD rule)
- LRT: $2 \Delta \text{NLL} \sim \chi_1^2$; 95% = 3.84; cutoff **1.92**
- 6 plots: [1.82, 4.61] (asymmetric!)
- 24 plots: [2.36, 3.75] ($\sqrt{4} = 2\times$ narrower)
- $\lambda = 1.5$ falsified; 2 and 4 survive

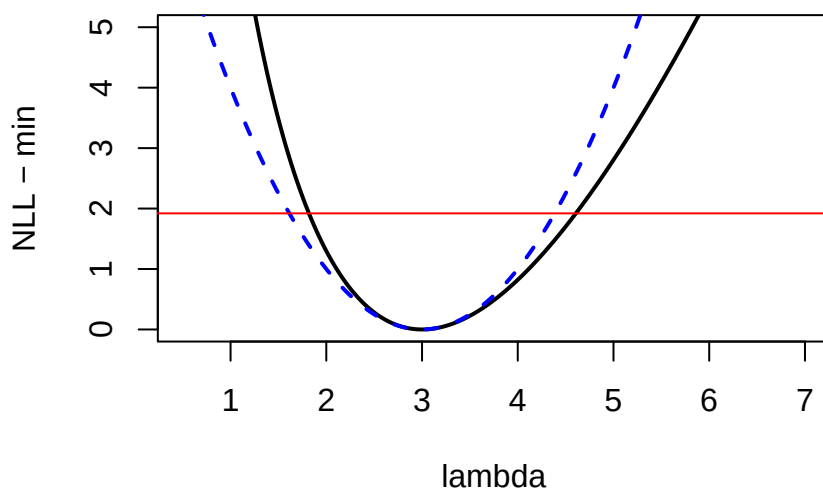
	95% interval
LR	[1.82, 4.61]
Bayes (flat)	[1.91, 4.74]

- Close numbers, different meaning: procedure vs. belief about λ

6. Quadratic approx. / Hessian (~6 min)

$$\begin{aligned} \text{NLL}(\lambda) &\approx \text{NLL}(\hat{\lambda}) + \frac{1}{2} \text{NLL}''(\hat{\lambda})(\lambda - \hat{\lambda})^2 \\ \Rightarrow \hat{\lambda} \pm \frac{1.96}{\sqrt{\text{NLL}''(\hat{\lambda})}}, \quad \text{SE} &= \frac{1}{\sqrt{\text{curvature}}} \end{aligned}$$

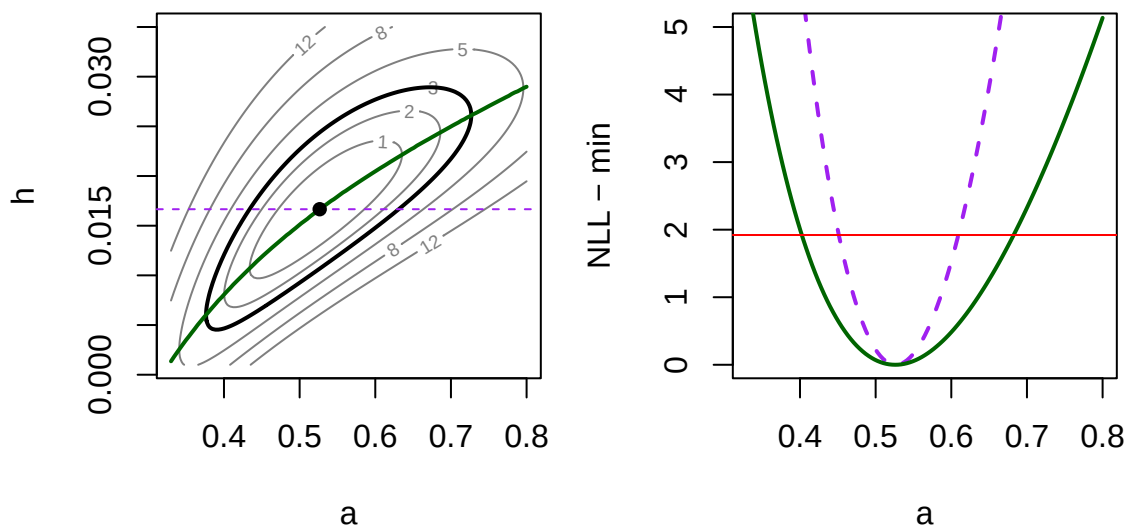
- Curvature = 2; SE = 0.71. **Don't derive** (Wed. take-home)
- `optim(..., hessian = TRUE)` gives 2 numerically
- Wald: [1.61, 4.39] vs LR [1.82, 4.61]



- **Sticking point:** why the *second* derivative? Sketch narrow vs wide parabola

7. More parameters (~4 min)

Bolker tadpoles, $p = a/(1 + ahN)$



- Tilted ellipse = correlated estimates ($\text{cor} \approx 0.82$)
- Slice (purple): other param fixed, too narrow
- Profile (green): re-optimize the others, correct
- Shortcut: `solve(hessian)` = var-cov matrix

8. Wrap (~1 min)

- Bottom of valley = estimate; width = uncertainty
- Same LRT next: COMPARE models

Discussion (~7 min)

1. “95% chance λ is in $[1.8, 4.6]$?” Which interval lets you say yes?
2. Is $\lambda = 1.5$ disproven? Is 3 proven?
3. $n = 6$: which interval do you trust most? How would you check?